Intelligent Customer Churn Prediction Using Hybrid Machine Learning and Explainable AI

PROJECT REVIEW 3 — METHODOLOGY VALIDATION, CODE IMPLEMENTATION & RESULTS

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PROJECT CONTEXT

Problem Statement and Review-3 Objectives

Customer churn is a binary classification problem with a minority churn class.

PROBLEM STATEMENT

Telecom customer churn reduces recurring revenue and increases replacement-acquisition costs. The project predicts which customers are likely to leave so retention teams can prioritize outreach. Because only about 26.5% of the sample churns, accuracy alone can hide missed churners and misleading risk scores.

BINARY CLASSIFICATION

CHURN = 1 | STAY = 0

Review-3 research question

Do hybrid classifiers offer a consistent advantage when discrimination, recall, calibration, threshold effects and paired uncertainty are considered together?

OBJECTIVES FOR THIS REVIEW

- Validate the data-preparation and evaluation workflow.
- Document implemented baseline and hybrid model pipelines.
- Compare test-set and five-fold cross-validation results.
- Explain threshold trade-offs and statistical uncertainty.
- Show dashboard features and distinguish tested from pending execution.
- Document repository organization and research-paper status.

DATASET AND DATA PREPARATION

IBM Telco Customer Churn Dataset

The supplied project uses the commonly circulated 7,043-row telco customer CSV.

CUSTOMER RECORDS	ORIGINAL COLUMNS	CHURNERS	CHURN RATE
7,043	21	1,869	26.54%

FEATURE GROUPS

Demographics	Gender, senior citizen, partner, dependents
Services	Internet, phone, security, support, streaming
Account	Tenure, contract, payment method, billing
Target	Churn: Yes / No converted to 1 / 0

DATA QUALITY CHECKS

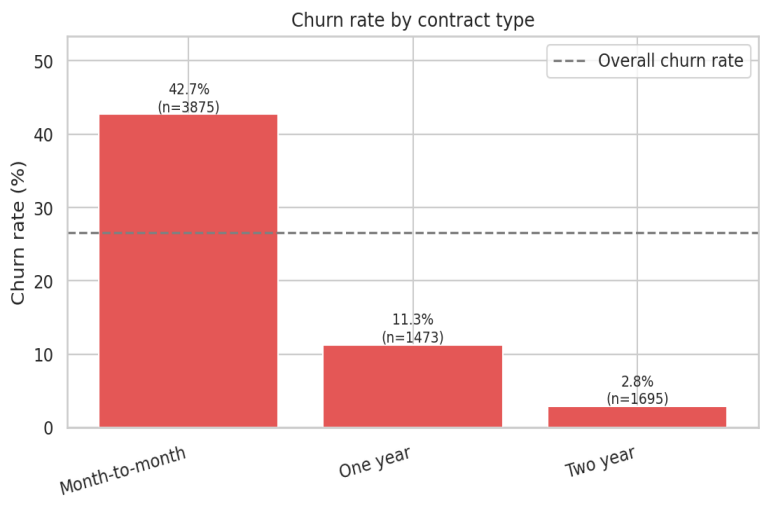
- 11 blank TotalCharges values converted to numeric 0.0; all have tenure = 0.
- 0 exact duplicate rows and 0 duplicate customer IDs.
- customerID excluded from model features; no IQR outlier rows removed.

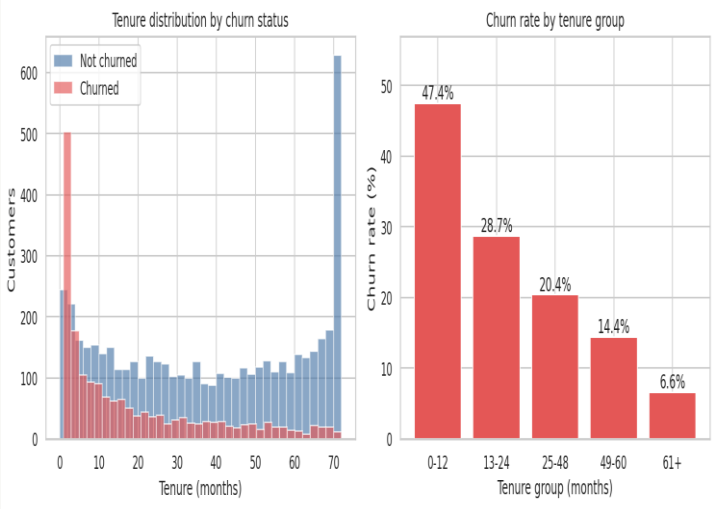
Stratified split: 5,634 training records and 1,409 test records (random_state = 42).

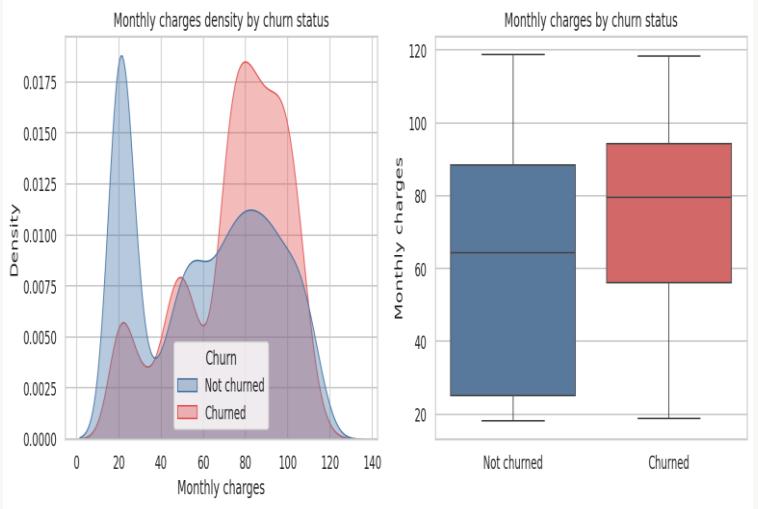
EXPLORATORY DATA ANALYSIS

Customer Segments Associated with Higher Churn

Descriptive patterns from the supplied analysis; these are associations, not causal effects.







Contract Month-to-month 42.7% • One year 11.3% • Two year 2.8%

Internet Fiber optic 41.9% • DSL 19.0% • No internet 7.4%

Payment Electronic check 45.3% • Other methods 15.2–19.1%

Tenure 0–12 months 47.4% • 61+ months 6.6%

METHODOLOGY VALIDATION

Leakage-Aware Experimental Protocol

The same train/test design is applied across the baseline and hybrid model experiments.

1 Raw data
CSV ingestion
7,043 × 21

2 Clean & derive
TotalCharges fix
12 engineered features

3 Split
Stratified 80/20
Seed 42

4 Preprocess
Impute • scale
One-hot encode

5 Train
Six baselines
Hybrid candidates

6 Evaluate
Holdout + 5-fold
training CV

VALIDATION CONTROLS

- Five-fold StratifiedKFold on training data only.
- Imputation, scaling and one-hot encoding learned within pipeline folds.

INTERPRETATION BOUNDARY

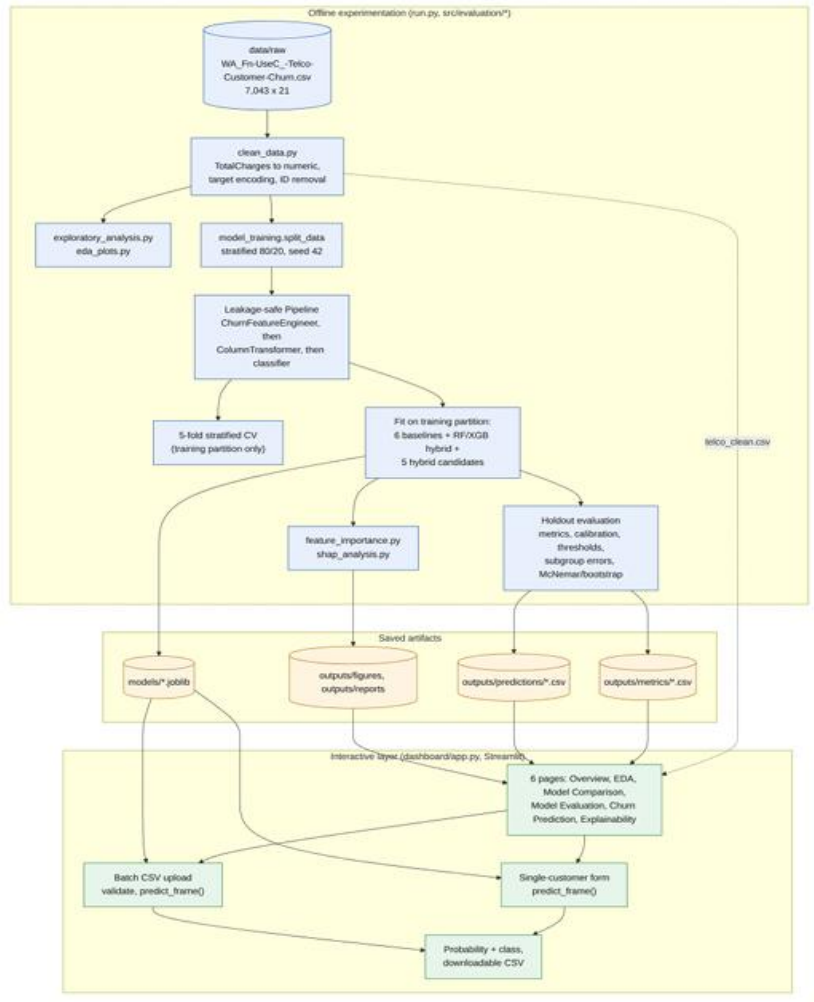
- The holdout was examined during candidate development; treat scores as exploratory.
- Use a fresh split or nested CV for confirmatory claims.

Full tuned-model results are not reported: the hyperparameter-search job did not complete.

SYSTEM DESIGN

Repository-Traced Implementation Architecture

Original architecture diagram from the project repository: offline experimentation, saved artifacts and interactive Streamlit inference.



1. OFFLINE EXPERIMENTATION

The raw Telco CSV is cleaned, explored and split using a stratified 80/20 design. A leakage-safe preprocessing pipeline feeds six baseline classifiers, the RF–XGBoost hybrid and five additional hybrid candidates. Five-fold stratified CV uses the training partition only.

2. SAVED ARTIFACTS

Trained model files, figures/reports, prediction CSV files and metric CSV files are saved separately. The dashboard consumes the persisted RF–XGBoost model path rather than retraining every candidate at request time.

3. INTERACTIVE STREAMLIT LAYER

Six dashboard pages expose Overview, EDA, Model Comparison, Model Evaluation, Churn Prediction and Explainability. Single-customer and batch CSV inputs are validated before the saved model returns churn probability, class and downloadable predictions.

Figure 1. Repository-traced architecture supplied by the project team. The diagram shows the offline training/evaluation path, persisted outputs and Streamlit prediction boundary.

MODERN TOOLS AND TECHNOLOGIES

Technology Stack Used in the Project

Tools are grouped by their function in the data, modelling, evaluation and delivery workflow.

PY

Python 3.12

Primary language and recommended environment

DATA

Pandas / NumPy

Data cleaning, tabular processing and numeric operations

SK

scikit-learn

Pipelines, preprocessing, classifiers, CV and metrics

XGB

XGBoost

Gradient-boosted trees and ensemble component

XAI

SHAP

Explainability module is present; generated report not verified

VIS

Plotly / Matplotlib / Seaborn

EDA, evaluation plots and dashboard visualisations

UI

Streamlit

Dashboard source and app entry points

OPS

joblib / Git / GitHub

Model persistence and repository versioning workflow

Environment noted in project documentation: Python 3.12; package and platform versions should be recorded with each reproducible run.

CODE IMPLEMENTATION

Implemented Data and Modelling Pipeline

The codebase separates ingestion, preprocessing, feature engineering, model training and evaluation.

DATA + FEATURE PIPELINE

Input	7,043 records • 21 columns • 19 original predictors
Cleaning	11 blank TotalCharges values → 0; drop customerID from predictors
Feature engineering	12 row-wise features: tenure group, service counts, charge features, contract/payment flags and interaction
Encoding	After one-hot encoding, model input contains 61 columns
Split	5,634 train • 1,409 test • stratified • seed 42

MODEL PIPELINES

01	Logistic Regression
02	Decision Tree
03	Random Forest
04	XGBoost
05	RBF Support Vector Machine
06	Artificial Neural Network (MLP)
07	**RF + XGBoost hybrid**

Five additional hybrid-candidate experiments are evaluated separately on the same saved holdout.

CODE VALIDATION

Execution Evidence and Verification Boundaries

Static checks and saved run reports support the implementation claims; skipped stages are identified explicitly.

VERIFIED IN THE CURRENT PROJECT ARCHIVE

PYTHON SOURCE FILES

37

compileall completed without errors

STATIC SYNTAX CHECK PASSED

- The project archive compiles with python -m compileall -q.
- Metric CSVs, model artifacts, evaluation figures and report files are present.

RECORDED EXECUTION / PENDING CHECKS

- A prior validation report records a successful run.py --skip-cv --skip-shap execution and separately completed five-fold CV.
- The --skip-cv --skip-shap run did not validate full CV or SHAP execution.
- Hyperparameter tuning did not complete; no tuned result is claimed.
- Streamlit launch and full SHAP artifact generation remain unverified in the audit.

Code syntax and saved outputs are verified; full end-to-end runtime evidence still requires a clean-environment rerun.

DASHBOARD EXECUTION

Six Functional Areas in the Streamlit Application

The dashboard source exposes analysis and prediction pages; execution status is reported conservatively.

01 Overview

Project summary and high-level churn profile

02 EDA

Contract, payment, internet, tenure and charge patterns

03 Model Comparison

Saved metric table and model rankings

04 Model Evaluation

Confusion matrix, ROC curve and PR curve

05 Churn Prediction

Single-customer form and batch CSV scoring

06 Explainability

Permutation importance, SHAP and tree-based importance if outputs exist

Original screenshots captured on 25-10-2026 show the Streamlit dashboard running locally at localhost:8501. A clean-environment rerun and formal usability study remain separate validation tasks.

Local launch command after installing requirements: `streamlit run app.py`

COMPARATIVE ANALYSIS

Baseline and Existing Hybrid Model Results

Fixed 20% holdout metrics from the supplied model_comparison.csv; positive class = churn.

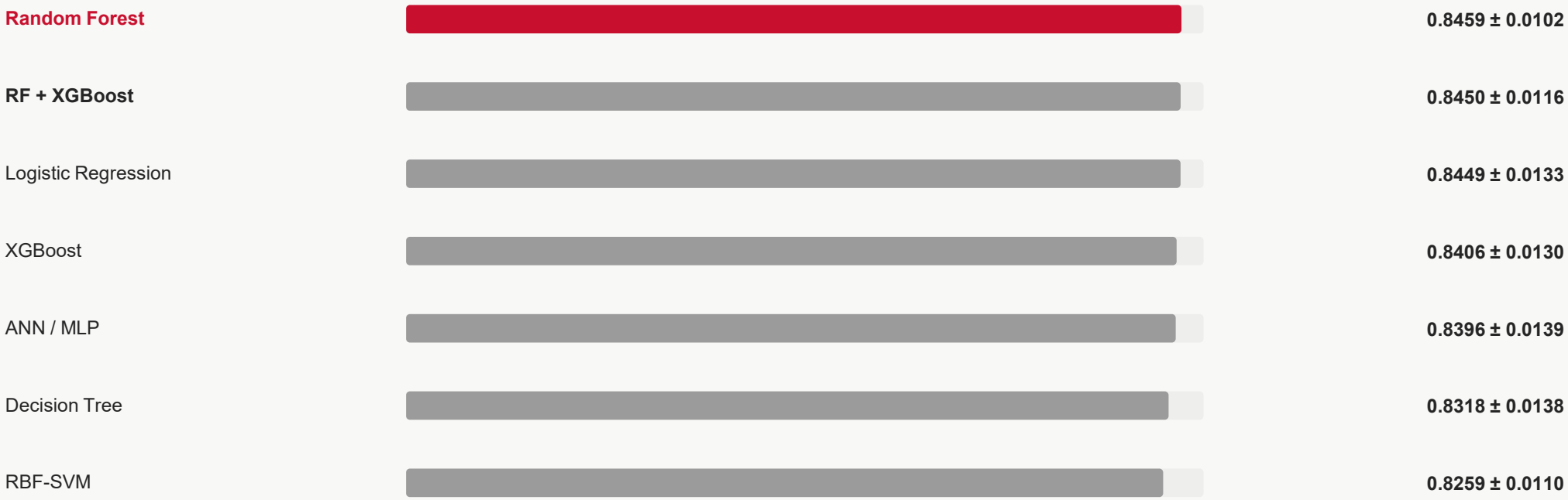
Model	Accuracy	Precision	Recall	F1	ROC-AUC	PR-AUC
Logistic Regression	0.7353	0.5008	0.7888	0.6127	0.8420	0.6370
Decision Tree	0.7488	0.5181	0.7674	0.6185	0.8321	0.6244
Random Forest	0.7693	0.5476	0.7540	0.6344	0.8413	0.6462
XGBoost	0.7530	0.5233	0.7807	0.6266	0.8428	0.6498
RBF-SVM	0.7828	0.5833	0.6364	0.6087	0.8249	0.6142
ANN / MLP	0.7977	0.6353	0.5588	0.5946	0.8365	0.6230
RF + XGBoost	0.7644	0.5396	0.7647	0.6327	0.8439	0.6510

Metric leaders differ: MLP has highest accuracy / precision; Logistic Regression has highest recall; RF + XGBoost has highest ROC-AUC and PR-AUC in this table.

METHODOLOGY VALIDATION

Five-Fold Cross-Validation on Training Data

Mean ROC-AUC ± standard deviation across five stratified folds. The holdout is not used in CV.



Random Forest has the highest CV mean ROC-AUC (0.8459); the RF–XGBoost hybrid is close (0.8450). The difference is smaller than fold variability.

PROPOSED HYBRID COMPARISON

Random Forest + XGBoost Weighted Soft Voting

P_hybrid = 0.5 × P_RF + 0.5 × P_XGB; classify as churn when probability ≥ 0.50.

WHY THIS HYBRID?

- Random Forest uses bagging across trees to reduce variance.
- XGBoost uses sequential boosting to correct residual errors.
- Averaging predicted probabilities creates one risk score.
- The hybrid was selected before results were inspected, as required by the project brief.

TEST-SET COMPARISON

Metric	RF	XGBoost	Hybrid
ROC-AUC	0.8413	0.8428	0.8439
PR-AUC	0.6462	0.6498	0.6510
F1	0.6344	0.6266	0.6327
Recall	0.7540	0.7807	0.7647
MCC	0.4850	0.4729	0.4822

The hybrid is competitive and has the highest ROC-AUC / PR-AUC in this 7-model table, but it does not lead every metric.

COMPARATIVE ANALYSIS

Additional Hybrid Candidate Experiments

Five additional candidate designs were evaluated; these are distinct from the deployed dashboard hybrid.

Candidate	F1	ROC-AUC	PR-AUC	CV ROC-AUC
Stacking (RF/XGB/SVM/LR)	0.6387	0.8419	0.6464	0.8455
Weighted RF/XGB/SVM vote	0.6253	0.8443	0.6573	0.8440
Chi-square + ensemble	0.6216	0.8342	0.6464	0.8402
RF feature selection + ANN	0.5714	0.8406	0.6379	0.8399
XGBoost leaf embedding + ANN	0.5662	0.8291	0.6206	0.8311

HIGHEST CANDIDATE ROC-AUC

0.8443

HIGHEST CANDIDATE PR-AUC

0.6573

HIGHEST CANDIDATE F1

0.6387

Weighted soft voting leads candidate PR-AUC; stacking leads candidate F1. No one candidate leads every metric.

STATISTICAL COMPARISON

Do Hybrid Score Differences Establish a Winner?

Paired comparisons are required because models are evaluated on the same customers.

PAIRED EVIDENCE

- Weighted vote vs RF+XGBoost: paired McNemar p = 1.000.
- Weighted-vote PR-AUC difference vs RF+XGBoost: +0.0063.
- 95% bootstrap CI: −0.0005 to +0.0130, which includes zero.
- Multiple-comparison adjustment does not establish a robust hybrid advantage.

WHAT THE RESULTS SUPPORT

The hybrid ensemble is a competitive model and improves slightly over some baselines on selected metrics. The evidence does not show a universal winner over Logistic Regression, Random Forest or XGBoost.

Report metric-by-metric trade-offs, cross-validation spread and uncertainty. Avoid a claim of statistical superiority.

PERFORMANCE EVALUATION

Threshold Selection Changes the Retention Trade-Off

Example using the existing RF–XGBoost hybrid on the previously examined holdout.

Threshold	Precision	Recall	Specificity	False negatives	False positives
0.50	0.5396	0.7647	0.7643	88	244
0.30	0.4499	0.8877	0.6077	42	406

RECALL AT 0.50

76.5%

RECALL AT 0.30

88.8%

FALSE POSITIVES AT 0.50

244

FALSE POSITIVES AT 0.30

406

Lowering the threshold catches more churners but increases false alarms and unnecessary retention outreach. Thresholds should be set with validation data and explicit business costs.

These holdout threshold results are exploratory because this test partition was inspected during candidate development.

LITERATURE MODELS VS PROPOSED HYBRID

How the Current Approach Relates to Recent Studies

Published scores come from different datasets and protocols; compare methods, not raw accuracy values.

Study / year	Method in literature	Relevance to this project
Beeharry & Fokone (2021)	Flexible voting ensemble; IBM and Duke telco data	Motivates hybrid classifier comparison and imbalance-aware testing.
Edwine et al. (2022)	Feature selection, undersampling and RF / SVM / KNN comparison	Supports class-imbalance handling and multiple evaluation metrics.
Poudel et al. (2024)	Tabular telecom classifiers with global/local SHAP interpretation	Connects churn prediction to explanations; SHAP output still needs verification here.
Manzoor et al. (2024)	Review of 212 churn-prediction articles	Highlights ensembles, interpretability and business-aligned evaluation.
Imani et al. (2025)	Systematic review of 2020–2024 ML/DL studies	Identifies imbalance, drift, explainability and practical metric issues.
Rasool et al. (2025)	Stacking ensemble with SHAP analysis	Direct methodological comparator for proposed RF–XGBoost hybrid.
El Attar & El-Hajj (2026)	Multi-model soft voting, SHAP and threshold optimisation	Recent ensemble/XAI direction; result values are not directly comparable.

Project positioning: compare six baseline classifiers and multiple hybrid candidates under one common benchmark protocol, adding CV, calibration/error measures and paired uncertainty.

EXPLAINABILITY AND CUSTOMER INSIGHTS

What the Current Analysis Can and Cannot Claim

Model-associated signals are useful for analysis, but do not establish causal effects.

EDA-BASED SIGNALS

Contract	Month-to-month churn = 42.7%
Tenure	Median: churners 10 months vs stayers 38
Internet	Fiber-optic churn = 41.9%
Payment	Electronic-check churn = 45.3%
Combined segment	Month-to-month + fiber optic = 54.6%

EXPLAINABILITY STATUS

- SHAP implementation module is present in the repository.
- The supplied local dashboard screenshot shows permutation importance, mean absolute SHAP bars and a SHAP summary plot.
- Month-to-month contract, tenure and internet-service indicators appear among the strongest displayed signals.
- These attributions describe model behaviour; they are not causal explanations of why a customer leaves.

The screenshot is preserved as original project evidence; reproducible regeneration and artifact versioning should be retained with the repository.

GITHUB REPOSITORY UPDATE

Project Structure and Version-Control Package

The supplied Stage-8 project archive contains the following organized source and output areas.

src/	data • eda • features • models • evaluation • explainability • visualization
dashboard/	Streamlit app and reusable data-access / prediction components
data/	raw CSV and processed train / test / cleaned datasets
models/	saved baseline estimators, RF–XGBoost hybrid and preprocessing pipeline
outputs/	figures • metrics • predictions • reports
docs/ + notebooks/	methodology • literature register • validation reports • paper draft • EDA/model notebooks
Root files	README.md • requirements.txt • run.py • app.py • .env.example

Remote GitHub push cannot be confirmed from the supplied ZIP: it contains no .git metadata or remote URL. Add the repository link and latest commit hash after pushing.

RESEARCH PAPER DRAFT COMPLETION

Manuscript Structure, Milestone and Submission Status

The Review-3 schedule supplied in the milestone image records 07-10-2026 as the draft-completion date.

PAPER TITLE

Comparative Evaluation of Hybrid Models for Telecom Customer Churn Prediction: Discrimination, Calibration, and Error Analysis

Draft structure

Abstract • Introduction • Related Work • Dataset & Preprocessing • Methods • Experimental Setup • Results • Discussion • Limitations • Conclusion • References • Reproduction Commands

STATUS IN THE SUPPLIED PROJECT PACKAGE

- Editable DOCX and Markdown draft are included in docs/.
- The Stage-8 report says the draft was prepared on 25-10-2026 and had not been submitted.
- Tuning and SHAP execution remain incomplete; the paper discloses these limitations.
- Verify author details, source metadata, figures and final citation style before submission.

Date reconciliation: the review milestone lists 07-10-2026, while the later Stage-8 draft report records 25-10-2026. Confirm the intended completion date in the final submission.

REVIEW-3 FINDINGS

Limitations and Next Actions

The next iteration should strengthen reproducibility and external validity before any operational claim.

CURRENT LIMITATIONS

- Only one public, static benchmark dataset.
- Holdout partition was inspected during candidate development.
- Hyperparameter tuning job did not complete.
- SHAP artifacts were not verified in the supplied audit.
- Streamlit launch was not verified because the dependency was absent.
- No cost matrix or retention-uplift experiment is available.

FUTURE WORK

- Complete training-only hyperparameter tuning.
- Use fresh external/temporal evaluation or nested CV.
- Calibrate probabilities and select threshold using retention costs.
- Verify SHAP and save generated explanation artifacts.
- Install dependencies in a clean environment and launch-test dashboard.
- Evaluate fairness, drift and model performance on further datasets.

ORIGINAL PROJECT SCREENSHOTS

Dashboard Overview and Contract Analysis

All screenshots on this slide were supplied from the running Customer Churn DAV application; no illustrative or AI-generated pictures are used.

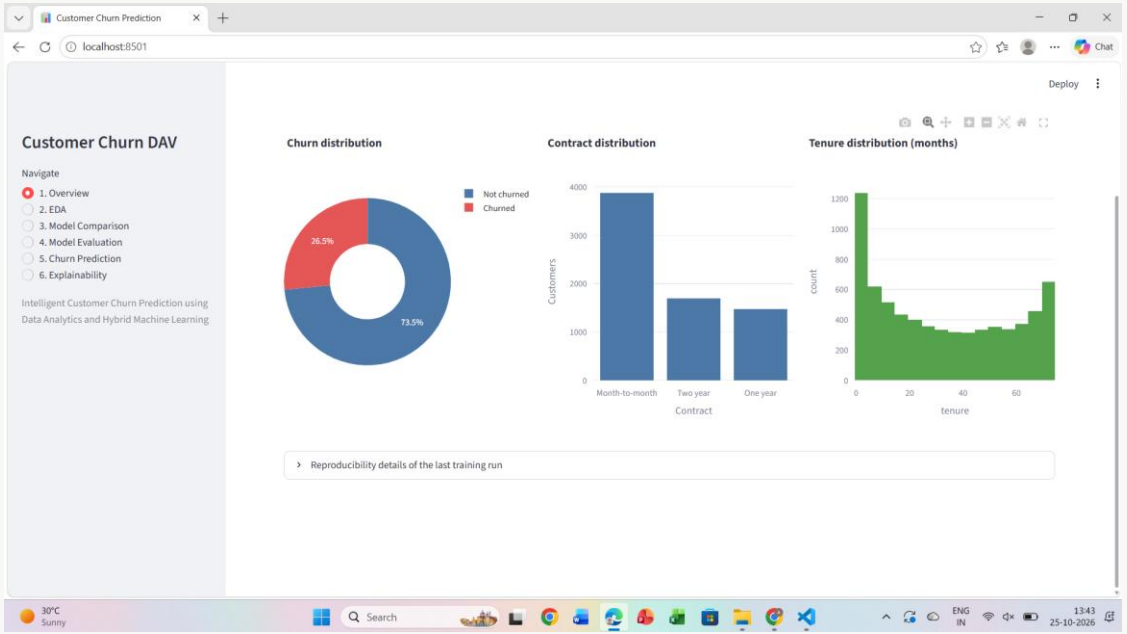


Figure 2. Dashboard overview: churn distribution, contract counts and tenure distribution.

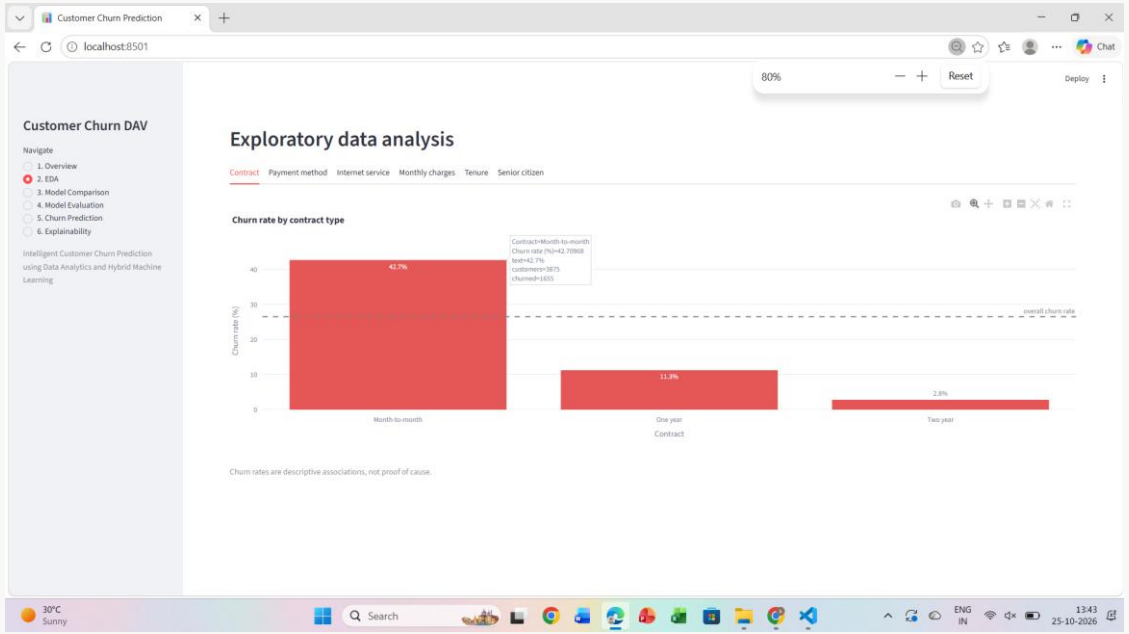


Figure 3. EDA contract analysis: month-to-month churn is visibly higher than one-year and two-year contracts.

Source: user-provided local application screenshots, captured 25 October 2026.

ORIGINAL PROJECT SCREENSHOTS

Monthly Charges and Customer Tenure

The images are unaltered captures of the dashboard EDA controls and plots.

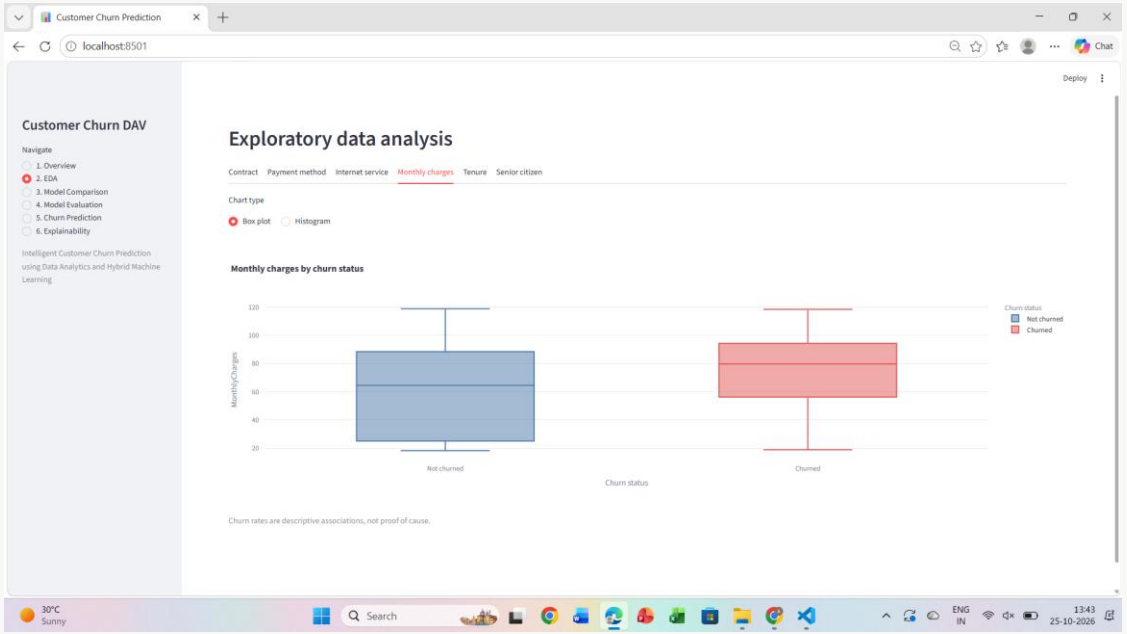


Figure 4. Monthly-charge box plot by churn status.

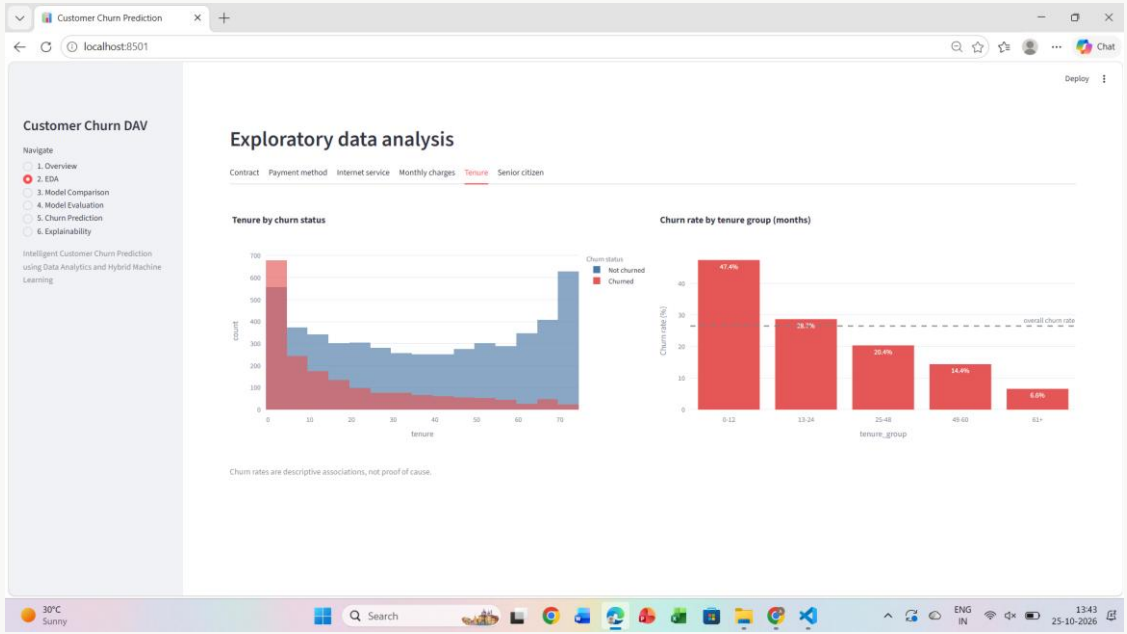


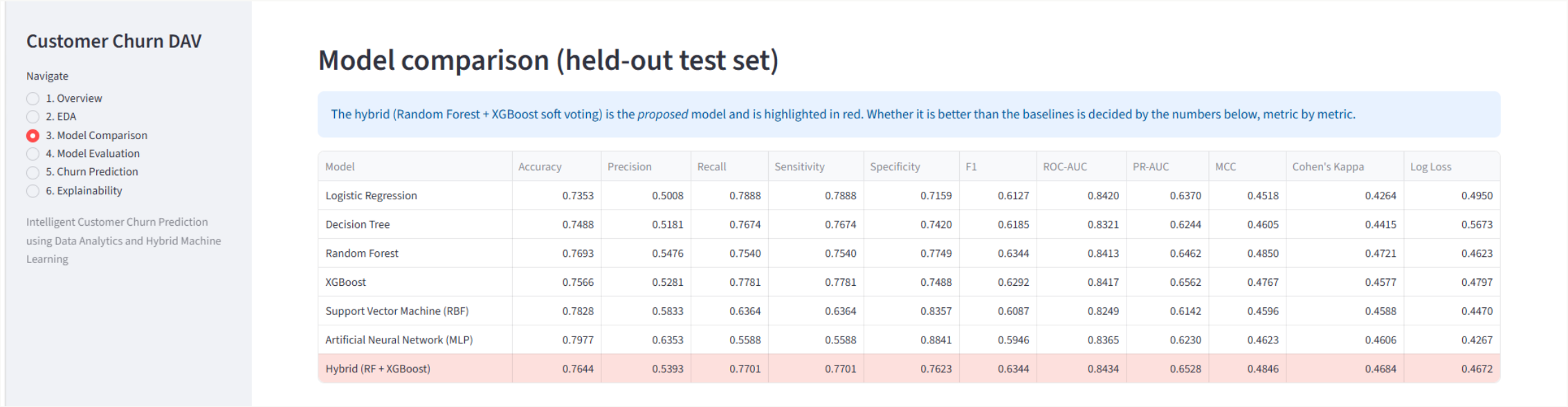
Figure 5. Tenure distribution and churn rate across tenure groups.

The displayed patterns are descriptive associations and should not be interpreted as causal effects.

ORIGINAL PROJECT SCREENSHOT

Model Comparison in the Streamlit Dashboard

The image below is the original dashboard capture with baseline and RF–XGBoost holdout metrics.



Model	Accuracy	Precision	Recall	Sensitivity	Specificity	F1	ROC-AUC	PR-AUC	MCC	Cohen's Kappa	Log Loss
Logistic Regression	0.7353	0.5008	0.7888	0.7888	0.7159	0.6127	0.8420	0.6370	0.4518	0.4264	0.4950
Decision Tree	0.7488	0.5181	0.7674	0.7674	0.7420	0.6185	0.8321	0.6244	0.4605	0.4415	0.5673
Random Forest	0.7693	0.5476	0.7540	0.7540	0.7749	0.6344	0.8413	0.6462	0.4850	0.4721	0.4623
XGBoost	0.7566	0.5281	0.7781	0.7781	0.7488	0.6292	0.8417	0.6562	0.4767	0.4577	0.4797
Support Vector Machine (RBF)	0.7828	0.5833	0.6364	0.6364	0.8357	0.6087	0.8249	0.6142	0.4596	0.4588	0.4470
Artificial Neural Network (MLP)	0.7977	0.6353	0.5588	0.5588	0.8841	0.5946	0.8365	0.6230	0.4623	0.4606	0.4267
Hybrid (RF + XGBoost)	0.7644	0.5393	0.7701	0.7701	0.7623	0.6344	0.8434	0.6528	0.4846	0.4684	0.4672

The dashboard compares accuracy, precision, recall, sensitivity, specificity, F1, ROC-AUC, PR-AUC, MCC, Cohen's kappa and log loss. In this captured table, the hybrid is strongest on ROC-AUC, but not on every metric.

Figure 6. Original model-comparison view captured from the running dashboard on 25 October 2026.

ORIGINAL PROJECT SCREENSHOTS

Confusion Matrix, ROC Curve and Precision–Recall Curve

Both images are direct screenshots of the dashboard Model Evaluation page.

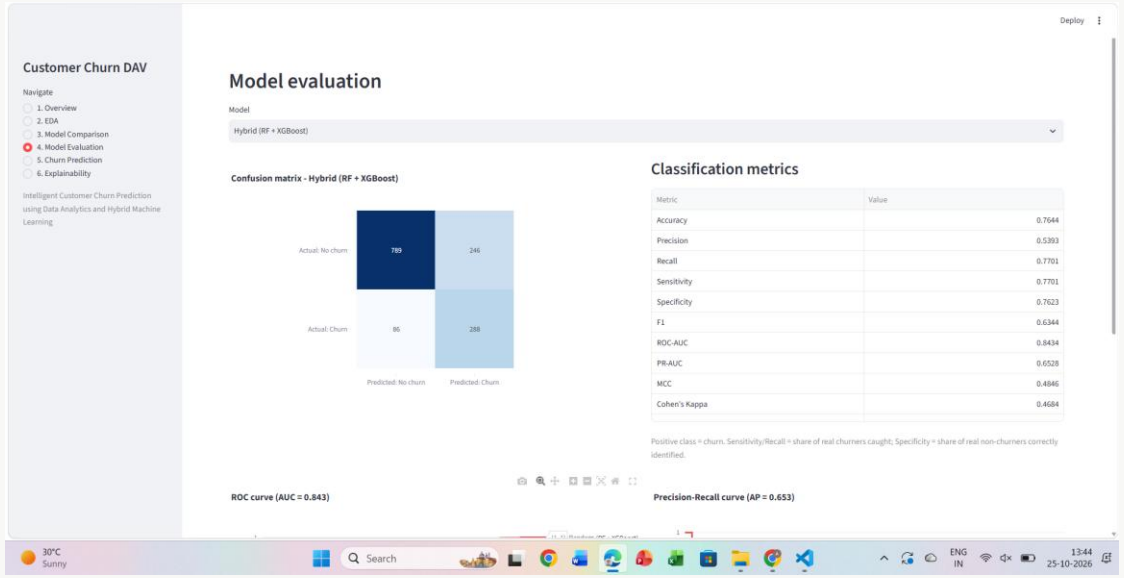


Figure 7. Confusion matrix and classification metrics for the RF–XGBoost hybrid.

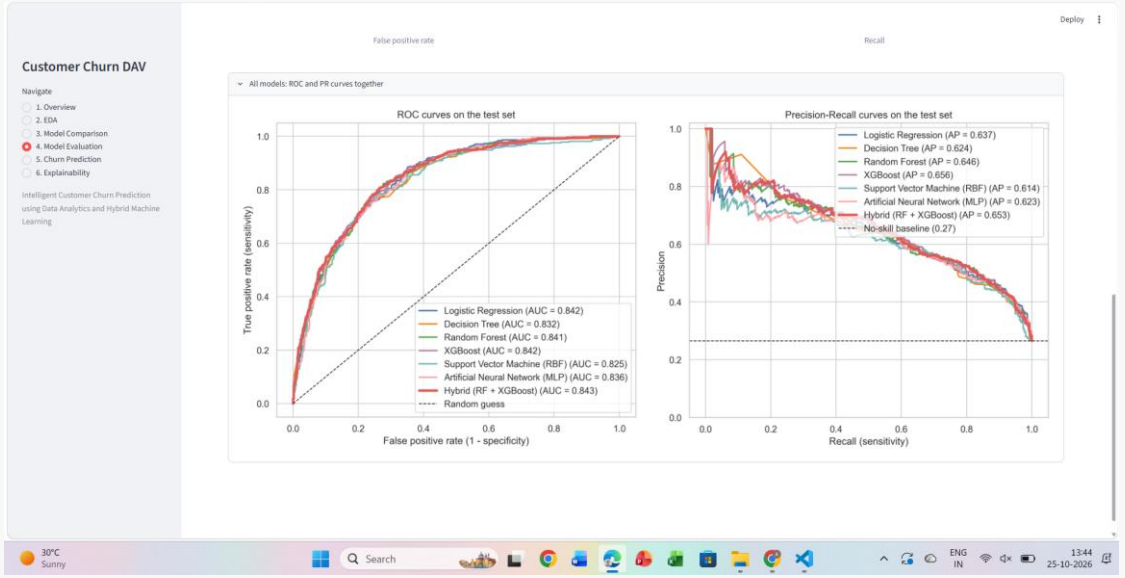


Figure 8. ROC and precision–recall curves for the compared classifiers.

The positive class is churn. ROC-AUC and PR-AUC summarize ranking quality; threshold-dependent metrics depend on the selected operating point.

ORIGINAL PROJECT SCREENSHOTS

Single-Customer Prediction and Explainability

The screen captures preserve the actual interface and explanation visuals produced by the project dashboard.

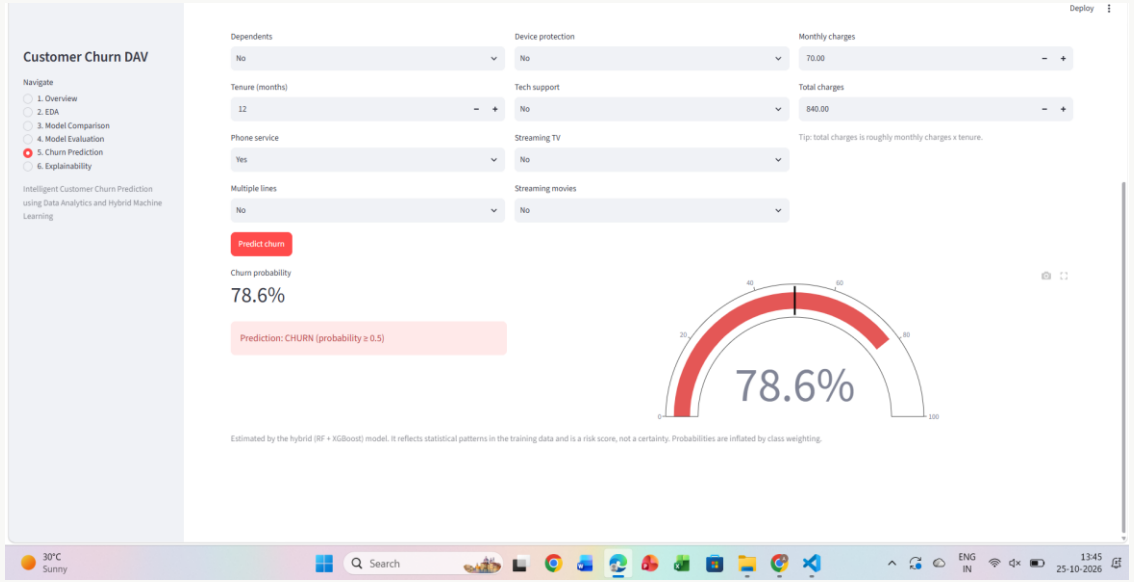


Figure 9. Single-customer form returning a 78.6% churn probability and CHURN label for the shown input.

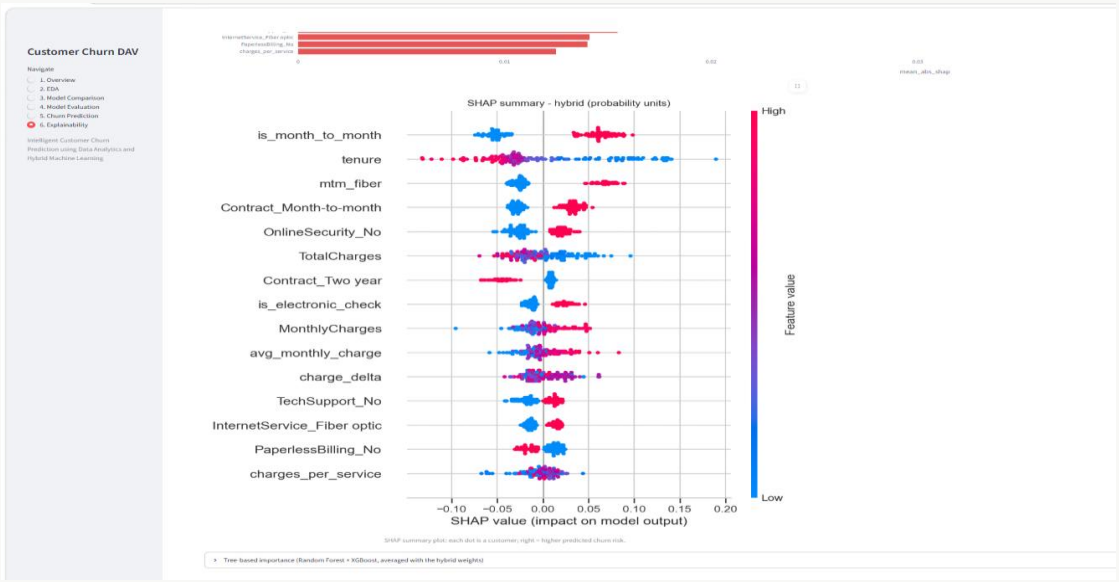


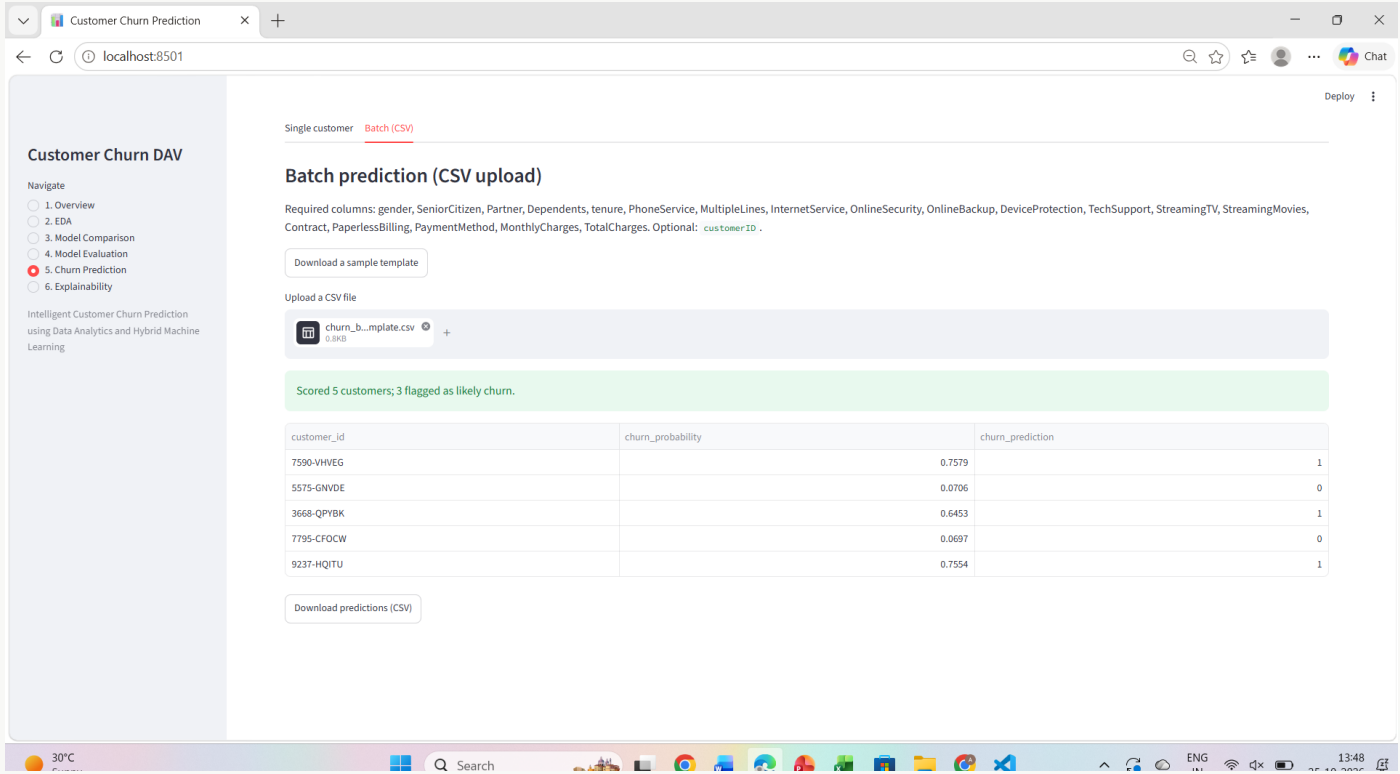
Figure 10. SHAP summary plot: feature effects are displayed in probability-output units for individual records.

A predicted probability is a model risk score, not a certainty. SHAP values explain model output and do not establish causation.

ORIGINAL PROJECT SCREENSHOTS

Batch Prediction and Downloadable Results

The screenshots show the project's CSV upload flow and the resulting per-customer probability and class output.



	A	B	C	D
1	customer_id	churn_probability	churn_prediction	
2	7590-VHVEG	0.757869351	1	
3	5575-GNVDE	0.070628411	0	
4	3668-QPYBK	0.645287768	1	
5	7795-CFOCW	0.069683905	0	
6	9237-HQITU	0.755388001	1	
7				
8				
9				
10				

Captured test batch: 5 customers scored; 3 were flagged as likely churn. The exported CSV contains customer_id, churn_probability and churn_prediction.

Figure 11. Batch CSV upload result in the dashboard. Figure 12. Original spreadsheet view of the downloaded prediction file.

Thank You